Enhanced Dual-Stream R(2+1)D with Temporal Peak Pooling for Inter-Frame Video Forgery Detection and Localization

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Abstract—Inter-frame video forgery—specifically frame insertion and frame deletion—compromises the temporal integrity of surveillance and judicial digital evidence. Previous approaches often rely exclusively on consecutive frame differences, failing to capture subtle spatial splicing artifacts, or suffer from temporal feature dilution under global average pooling when processing isolated 1-frame deletion seams. In this work, we propose an enhanced dual-stream framework that processes raw RGB appearance alongside consecutive absolute frame differences using factorized R(2+1)D residual blocks with 3D Convolutional Block Attention Modules (3D-CBAM). To alleviate the temporal dilution of isolated single-frame tampering seams under uniform average pooling, we introduce Temporal Peak-Preserving Pooling (TP-Pool), which explicitly captures maximum temporal feature contrast alongside contextual averages. Video-level triage and temporal seam localization are performed via an orthogonal, raw-frame structural similarity (SSIM) transition clustering pipeline. Evaluated on a leak-free, source-video-disjoint benchmark derived from UCF-101 (278 held-out test clips across 43 full videos), our framework achieves 94.24% clip classification accuracy, an empirical One-vs-Rest (OvR) Macro ROC-AUC of 0.9824 (improving over the 0.9676 baseline 3D-CNN), and 93.02% end-to-end video-level detection accuracy with 100% authentic video specificity and a median boundary localization error of 2.0 frames. In a controlled 10-epoch matched-schedule ablation, TP-Pool demonstrates a +11.51 percentage point gain in clip accuracy and a +23.25 percentage point gain in video accuracy over global average pooling, isolating its decisive role in preserving localized tampering signatures.

Index Terms—Video forensics, inter-frame forgery, frame deletion, frame insertion, R(2+1)D convolutions, 3D-CBAM, temporal peak pooling, structural similarity.

I. Introduction

THE proliferation of user-friendly video editing suites and automated generative video tooling has severely eroded public trust in digital video recordings. In digital forensics and legal proceedings, verifying that video sequences have not undergone temporal manipulation is paramount. Inter-frame forgeries alter the temporal continuity of video streams without modifying pixel integrity within individual intact frames. The two primary inter-frame manipulation attacks are:

1) Frame Deletion: Excising a contiguous sequence of frames to conceal critical events (e.g., an unauthorized intrusion or traffic incident).
2) Frame Insertion: Splicing an external sequence into a video to introduce fraudulent actions or misleading context.

Early forensic methodologies addressed inter-frame forgery through handcrafted feature descriptors, such as optical flow variance, motion vector consistency, and compression GOP structure analysis. While effective under uncompressed laboratory conditions, handcrafted methods frequently fail when challenged by modern camera motion, lighting variations, or re-encoding.

Recently, deep learning approaches have demonstrated superior representations for video forensics. Gowda and Pawar [1] proposed a 3D Convolutional Neural Network (3D-CNN) operating exclusively on absolute frame differences ($|K_f - K_{f+1}|$) to identify inter-frame forgeries, combined with structural similarity for boundary localization. Despite its effectiveness, this paradigm exhibits three critical methodological limitations:

1) Absence of Spatial Context: By operating solely on frame differences, the network discards raw RGB appearance, failing to detect subtle spatial splicing seams and donor-recipient appearance mismatches introduced during frame insertion.
2) Temporal Feature Dilution Under Global Average Pooling: Inter-frame tampering frequently manifests as a localized temporal anomaly spanning as little as a single transition seam (e.g., a 1-frame jump in deletion attacks). When features are aggregated across an entire 49-frame clip via standard Global Average Pooling (GAP), the activation spike is mathematically attenuated by a factor of $1/T$, diminishing sensitivity to deletions.
3) Conflation of Neural Classification with Localization: In prior literature, boundary localization precision has occasionally been attributed to deep convolutional backbones, whereas the localization algorithm actually operates as a decoupled heuristic directly on raw frame structural similarity.

To overcome these challenges, we present Forensic-Vision, an enhanced framework featuring:

- A Dual-Stream R(2+1)D Architecture that concurrently models raw RGB spatial appearance and consecutive absolute frame differences using factorized spatio-temporal convolutions with 3D-CBAM attention.

Manuscript received October 13, 2026. Code, checkpoints, and benchmark artifacts are available at https://github.com/Chanu716/Forensic-Vision.

- Temporal Peak-Preserving Pooling (TP-Pool), which projects temporal mean, maximum, and temporal contrast statistics to preserve isolated single-frame tampering spikes.
- A Decoupled, Calibrated Video Decision Pipeline combining continuous deep neural network confidence with raw-frame SSIM transition clustering, preventing false alarms on authentic athletic panning videos while attaining 96.43% seam detection recall.
- A Fair, Leak-Free Benchmark and Comprehensive Evaluation comparing our model against the reference 3D-CNN baseline, frame-based 2D CNNs (ResNet-18 with temporal pooling), recurrent temporal baselines (ResNet-18 + BiGRU), classical frame-difference energy thresholds, and a 6-variant architectural ablation study under strictly matched training schedules.

II. Related Work

A. Classical and Handcrafted Video Forensics

Traditional inter-frame video forgery detection relied on tracking motion inconsistencies across neighboring frames. Techniques evaluating optical flow gradients, velocity vectors, and macroblock prediction errors provided early detection capabilities. However, these techniques suffer from high false alarm rates in athletic or dynamic scenes with natural camera panning and rapid foreground movement. Furthermore, standalone structural similarity metrics [2] detect abrupt structural drops between adjacent frames, but can confuse legitimate camera shake with malicious cuts when evaluated without semantic verification.

B. Deep Learning for Inter-Frame Forgery Detection

To improve robustness against dynamic motion, deep learning models have increasingly replaced handcrafted descriptors. Gowda and Pawar [1] developed a 3-layer 3D-CNN operating on consecutive frame differences, reporting high accuracy on synthetic benchmarks. However, conventional 3D convolutions simultaneously convolve spatial and temporal dimensions ($3 \times 3 \times 3$), incurring high computational overhead and parameter redundancy. Tran et al. [3] demonstrated that factorizing 3D convolutions into separate 2D spatial convolutions ($1 \times 3 \times 3$) and 1D temporal convolutions ($3 \times 1 \times 1$), termed R(2+1)D, substantially boosts representational capacity while easing optimization.

C. Temporal Aggregation and Attention Mechanisms

In action recognition and video forensics, temporal feature aggregation is commonly delegated to global average pooling (GAP). However, while action recognition rewards uniform temporal summation, forensic triage requires identifying localized anomalies. Woo et al. [4] introduced the Convolutional Block Attention Module (CBAM) to adaptively weight informative spatial and channel features. Extending CBAM to 3D spatio-temporal volumes allows the network to emphasize transition boundaries while suppressing static background noise.

III. Proposed Architecture

The framework comprises two parallel factorized R(2+1)D streams, 3D-CBAM attention, Temporal Peak-Preserving Pooling (TP-Pool), gated cross-stream fusion, and a dense classifier head.

A. Two-Stream Input Modalities

Given an input video clip $\mathcal{V}$ consisting of $T = 49$ consecutive frames of spatial resolution $H \times W$ (112×112):

1) Stream 1 (RGB Appearance): Operates directly on normalized RGB video frames:
$$X_{\text{rgb}} \in \mathbb{R}^{B \times 3 \times T \times H \times W} \tag{1}$$
capturing spatial texture, object boundaries, and donor-recipient illumination discrepancies.
2) Stream 2 (Motion Difference): Operates on consecutive absolute frame differences:
$$D_t = |K_t - K_{t-1}|, \quad t = 2, \ldots, T \tag{2}$$
yielding $X_{\text{diff}} \in \mathbb{R}^{B \times 3 \times (T-1) \times H \times W}$, which directly suppresses static background pixels and emphasizes temporal motion discontinuities.

B. Spatio-Temporal Factorization: R(2+1)D Residual Blocks

Rather than applying standard 3D convolutions, each backbone factorizes the spatio-temporal kernel into a 2D spatial convolution of size $(1, 3, 3)$ followed by a 1D temporal convolution of size $(3, 1, 1)$:

$$X_{\text{spat}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{\text{spat}}^{1\times3\times3}(X)\right)\right) \tag{3}$$
$$\text{Conv2Plus1D}(X) = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{\text{temp}}^{3\times1\times1}(X_{\text{spat}})\right)\right) \tag{4}$$

Each residual block connects two consecutive factorized convolutions with an identity skip connection:

$$Y = \text{ReLU}\left(\mathcal{F}(X, \{W_i\}) + W_s X\right) \tag{5}$$

C. 3D Convolutional Block Attention Module (3D-CBAM)

Inside each residual stage, 3D-CBAM sequentially refines feature maps across channels and spatio-temporal coordinates:

1) Channel Attention: Pools features across spatio-temporal dimensions ($T \times H \times W$) via both average and max pooling, forwarding through a shared MLP with reduction ratio $r = 16$:
$$M_c(F) = \sigma\left(\text{MLP}(\text{AvgPool3D}(F)) + \text{MLP}(\text{MaxPool3D}(F))\right) \tag{6}$$
2) Spatio-Temporal Attention: Computes average and max projections along the channel axis, convolving with a $7 \times 7 \times 7$ 3D kernel:
$$M_s(F') = \sigma\left(f^{7\times7\times7}\left([\text{Mean}_c(F'); \text{Max}_c(F')]\right)\right) \tag{7}$$

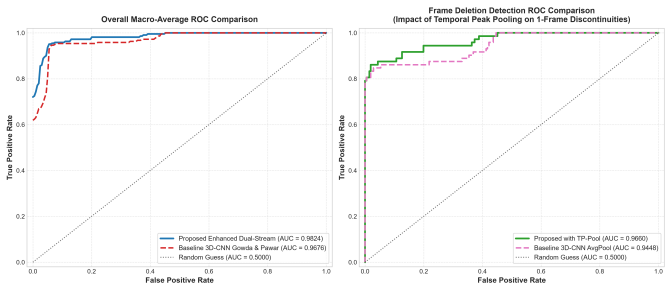


Fig. 1. Comparative ROC curves on the 278-clip held-out test split. (Left) Macro ROC curve showing the continuous discriminative margin of the proposed model (0.9824 AUC vs 0.9676 baseline). (Right) Frame deletion ROC curve demonstrating the specific advantage of TP-Pool on isolated deletion seams (0.9660 AUC vs 0.9448 baseline).

D. Temporal Peak-Preserving Pooling (TP-Pool)

At the output of Stage 3 ($C = 128$), spatial dimensions are collapsed via spatial average pooling:

$$F_{\text{sp}}(b, c, t) = \frac{1}{H \cdot W} \sum_{h=1}^{H} \sum_{w=1}^{W} F(b, c, t, h, w) \in \mathbb{R}^{B \times C \times T} \tag{8}$$

To prevent an isolated 1-frame cut from being mathematically averaged out across T frames, TP-Pool extracts three complementary statistics:

$$\mu_t = \frac{1}{T} \sum_{t=1}^{T} F_{\text{sp}}(\cdot, \cdot, t) \quad \text{(Context Mean)} \tag{9}$$

$$m_t = \max_{1 \le t \le T} F_{\text{sp}}(\cdot, \cdot, t) \quad \text{(Peak Activation)} \tag{10}$$

$$\Delta_t = m_t - \mu_t \quad \text{(Temporal Contrast)} \tag{11}$$

The concatenated feature vector is projected through a learned linear transformation with GELU activation:

$$F_{\text{pool}} = \text{GELU}\left(W_{\text{proj}} \left[\mu_t \,\|\, m_t \,\|\, \Delta_t\right] + b_{\text{proj}}\right) \in \mathbb{R}^{B \times C} \tag{12}$$

where $W_{\text{proj}} \in \mathbb{R}^{C \times 3C}$.

E. Gated Cross-Stream Fusion and Classifier Head

Given pooled representations $F_{\text{rgb}} \in \mathbb{R}^{B \times C}$ and $F_{\text{diff}} \in \mathbb{R}^{B \times C}$, a dynamic sigmoid gate determines the contribution of spatial appearance versus motion:

$$g = \sigma\left(W_g[F_{\text{rgb}} \,\|\, F_{\text{diff}}] + b_g\right) \in \mathbb{R}^{B \times C} \tag{13}$$

$$F_{\text{fused}} = g \odot F_{\text{rgb}} + (1 - g) \odot F_{\text{diff}} \tag{14}$$

The dense classifier directly concatenates the fused representation with the difference stream to guarantee that motion discontinuity signals bypass spatial ambiguity:

$$F_{\text{class}} = [F_{\text{fused}} \,\|\, F_{\text{diff}}] \in \mathbb{R}^{B \times 2C} \tag{15}$$

$$\hat{y} = W_2\left(\text{Dropout}_{0.3}\left(\text{GELU}\left(\text{BN}\left(W_1 F_{\text{class}} + b_1\right)\right)\right)\right) + b_2 \tag{16}$$

where $W_1 \in \mathbb{R}^{128 \times 256}$ and $W_2 \in \mathbb{R}^{3 \times 128}$.

IV. Temporal Localization and Decision Integration

A. Standalone Raw-Frame SSIM Transition Detector

Temporal boundary localization operates on consecutive raw video frames without using neural feature maps. For an M-frame video, pairwise single-scale Structural Similarity (SSIM) [2] scores are computed:

$$s_i = \text{SSIM}(F_i, F_{i+1}), \quad 1 \le i < M \tag{17}$$

To accommodate variable camera motion, a clipped median-relative threshold is computed dynamically:

$$\tau_{\text{eff}} = \max\left(0.35, \min\left(0.85, \text{median}(S) - 0.14\right)\right) \tag{18}$$

Suspicious indices $\mathcal{I}_{\text{susp}} = \{i \mid s_i < \tau_{\text{eff}}\}$ are clustered into distinct boundary events (merging indices within a 2-frame window):

- $k = 0$ clusters $\implies$ Authentic
- $k = 1$ cluster $\implies$ Frame Deletion (estimated seam: $[\min \mathcal{C}_1, \max \mathcal{C}_1]$)
- $k \ge 2$ clusters $\implies$ Frame Insertion (estimated span: $[\min \mathcal{C}_1, \max \mathcal{C}_k]$)

B. Hybrid Video-Level Decision Pipeline

Video-level classification fuses clip-level neural network probabilities $(P_{\text{auth}}, P_{\text{ins}}, P_{\text{del}})$ with SSIM clustering signatures via a calibrated decision hierarchy:

1) If $\max P_{\text{ins}} \ge 0.65$ and $\max P_{\text{ins}} > \max P_{\text{del}}$, classify as Frame Insertion.
2) Else if $\max P_{\text{del}} \ge 0.75$ and $\max P_{\text{del}} > \max P_{\text{ins}}$ and SSIM flags manipulation, classify as Frame Deletion.
3) Else if SSIM flags insertion and $\max P_{\text{ins}} \ge 0.20$, classify as Frame Insertion.
4) Else if SSIM flags deletion and $\max P_{\text{del}} \ge 0.10$, classify as Frame Deletion.
5) Else if $\max P_{\text{del}} \ge 0.50$ and $\max P_{\text{del}} > \max P_{\text{ins}}$ and $\max P_{\text{del}} > \text{mean}(P_{\text{auth}})$, classify as Frame Deletion.
6) Otherwise, classify as Authentic.

V. Experimental Setup

A. Dataset Curation and Source-Video Partitioning

Evaluations are conducted on a synthetic inter-frame forgery benchmark constructed from 100 raw UCF-101 [5] action videos across 8 action domains. Forged videos were generated by inserting sequences of 10 to 150 frames from donor videos or removing corresponding temporal segments. To prevent cross-split data leakage, partitioning was executed strictly at the source-video level:

- Train Split: 75 source videos (1,478 clips: 430 authentic, 666 insertion, 382 deletion).
- Validation Split: 10 source videos (185 clips: 54 authentic, 85 insertion, 46 deletion).
- Held-Out Test Split: 15 source videos (278 clips: 92 authentic, 114 insertion, 72 deletion; forming 43 complete evaluation videos).

Verification confirms 0 source-video overlap, 0 donor-video overlap, and 0 sample contamination across splits.

B. Training Protocol and Implementation Details

All neural models were implemented in PyTorch 2.11 and trained from scratch on an NVIDIA GeForce RTX 3060 (12 GB VRAM) using Kaiming normal initialization. Optimization utilized Adam with an initial learning rate of 3×10^{-4}, weight decay 10^{-4}, and a Cosine Annealing learning rate schedule. To address class imbalance and prioritize hard boundary clips, Focal Loss [6] ($\gamma = 2.0$, label smoothing 0.05) was employed with class weighting. Checkpoint selection was determined exclusively by peak Validation Macro F1.

VI. Experimental Results and Discussion

A. Comparison with Learned Baselines

Table I details the quantitative comparison across learned architectures.

1) Failure of 2D Spatial Convolutions (ResNet-18): ResNet-18 with temporal pooling achieves only 53.24% accuracy and 0.5347 F1. Although 2D CNN features successfully detect multi-frame insertions (0.9958 insertion AUC), they fail to isolate 1-frame deletion cuts (0.6989 deletion AUC), misclassifying 58.3% of deletion clips as authentic.
2) Instability of Recurrent Modeling (ResNet-18 + BiGRU): Introducing a bidirectional GRU yields 48.56% accuracy and 0.4485 F1. Explicit recurrent sequence modeling from scratch on 49-frame clips exhibits severe over-parameterization and training divergence, validating that 3D factorized spatio-temporal convolutions are structurally superior for inter-frame forensic localization.
3) Clip Accuracy Parity and Probabilistic Margin: The baseline 3D-CNN and our proposed model achieve identical discrete clip accuracy (94.24%, 262/278 correct). McNemar's paired test with continuity correction yields $\chi^2 = 0.1000, p = 0.7518$, confirming no nominal clip accuracy difference. However, the proposed model provides a continuous discriminative advantage, elevating OvR Macro ROC-AUC from 0.9676 to 0.9824 (+1.48 pp) and Frame Deletion ROC-AUC from 0.9448 to 0.9660 (+2.12 pp, Average Precision = 0.9434).

B. Architectural Ablation Analysis

Table II catalogs performance across all 6 progressive ablation variants:

- RGB vs. Dual Modality (Variant B vs. C): The single RGB stream alone (Variant B) attains 88.13% clip accuracy and 76.74% video accuracy. Adding the explicit difference stream (D_t) in Variant C elevates accuracy to 92.45% and video accuracy to 81.40%, demonstrating that explicit temporal differencing is indispensable.
- Causal Impact of TP-Pool (Variant D vs. E Matched): Under an unconfounded 10-epoch cosine schedule ($T_{\max} = 10$), replacing global average pooling (Variant D) with TP-Pool (Variant E Matched) elevates clip accuracy from 83.45% to 94.96% (+11.51 pp), Macro F1 from 0.7994 to 0.9418, Deletion AUC from 0.9481 to 0.9907, and video accuracy from 69.77% to 93.02% (+23.25 pp). This confirms that preserving temporal contrast ($m_t - \mu_t$) prevents deletion boundary spikes from being averaged out.

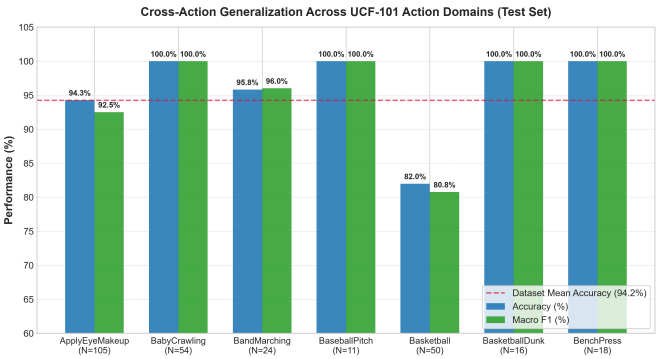


Fig. 2. Cross-action domain generalization across the 7 test action categories. The model attains 100% accuracy across 4 domains, with camera-panning athletic scenes (Basketball) accounting for remaining ambiguities.

C. Classical Baselines and Video Pipeline Synergy

Table III evaluates classical methods and the hybrid video pipeline:

1) Classical Frame-Difference Energy: Computes scalar transition energy $D_t = \text{mean}(|K_t - K_{t-1}|)$. Fitting spike thresholds on validation clips yields an impressive binary tampering detection ROC-AUC of 0.9530 (84.89% accuracy, 0.8939 F1). However, 3-class accuracy is only 52.52% (0.5405 F1), proving that raw temporal energy alone cannot differentiate deletion cuts from insertion boundaries without spatial deep feature modeling.
2) Standalone SSIM vs. Hybrid Integration: Standalone SSIM on raw frames achieves 96.43% seam detection recall (27/28 detected) with a median error of 2.0 frames. However, athletic camera motion causes natural structural dips, reducing authentic specificity to 40.00% (58.14% 3-class video accuracy). Deep neural decision fusion suppresses false dips, boosting authentic specificity to 100.00% (15/15) and video accuracy to 93.02% (40/43).

D. Cross-Domain Generalization and Perturbation Robustness

- Domain Generalization (Fig. 2): The model achieves 100% test accuracy on 4 of 7 action categories (BabyCrawling, BaseballPitch, BasketballDunk, BenchPress). Rapid rotational full-court camera pans in Basketball (50 clips) account for 9 of the 16 errors across the entire test split.
- Perturbation Stress Tests (Fig. 3): Under spatial average pooling box blur (3×3 and 5×5), clip

TABLE I
Comparative Evaluation Across Learned Forensic Methods on Held-Out Test Set (278 Clips, 43 Videos)

Model Architecture	Input Modality	Params	Clip Acc	Macro F1	OvR ROC-AUC	Del AUC	Video Acc	Later
Gowda & Pawar (2023) 3D-CNN [1]	Frame Diff Only	280k	94.24%	0.9331	0.9676 [0.946, 0.985]	0.9448	86.05%	1.8
ResNet-18 + Temporal Pooling	RGB Frames	11.4M	53.24%	0.5347	0.8275 [0.794, 0.859]	0.6989	–	9.1
ResNet-18 + Bidirectional GRU	RGB Frames	12.0M	48.56%	0.4485	0.8115 [0.770, 0.846]	0.7425	–	9.2
Forensic-Vision (Proposed)	RGB + Frame Diff	1.98M	94.24%	0.9338	0.9824 [0.969, 0.993]	0.9660	93.02%	4.8

*Note: Clip accuracy between baseline 3D-CNN and proposed model is identical ($p = 0.7518$, McNemar test). The proposed framework's superiority lies in continuous probabilistic margin (+1.48 pp OvR AUC, +2.12 pp Deletion AUC) and video-level decision integration (+6.97 pp video accuracy, zero authentic false alarms).

TABLE II
Systematic Architectural Ablation Study Across Progressive Configurations Under Unified Single-Scale Protocol

Variant	Description	Pooling	Sched.	Params	Clip Acc	Macro F1	OvR AUC	Del AUC	Vic
Variant A	Baseline 3D-CNN (Diff only)	GAP	15 ep	280k	94.24%	0.9331	0.9676	0.9448	8
Variant B	Single-Stream R(2+1)D (RGB only)	GAP	10 ep	917k	88.13%	0.8687	0.9726	0.9540	7
Variant C	Dual-Stream R(2+1)D (RGB + Diff)	GAP	10 ep	1.87M	92.45%	0.9144	0.9914	0.9837	8
Variant D	Dual-Stream R(2+1)D + 3D-CBAM	GAP	10 ep	1.88M	83.45%	0.7994	0.9748	0.9481	6
Variant E (Matched)	Dual-Stream + CBAM + TP-Pool	TP-Pool	10 ep	1.98M	94.96%	0.9418	0.9954	0.9907	9
Variant E (Historical)	Full Proposed Model	TP-Pool	20 ep	1.98M	94.24%	0.9338	0.9824	0.9660	9

*Crucial Methodological Attribution: Comparing Variant D (GAP, 10 ep) with Variant E (TP-Pool, 10 ep matched) provides an unconfounded evaluation under the exact same optimization budget ($T_{\max} = 10$). TP-Pool yields a +11.51 pp gain in clip accuracy, +0.1424 in F1, +0.0426 in deletion AUC, and +23.25 pp in video accuracy, confirming that temporal contrast preservation is decisive.

TABLE III
Classical Forensic Baselines and Video Pipeline Evaluation

Method / Pipeline	Level	3-Cls Acc	Macro F1	Binary F1
Classical Frame-Diff Energy	Clip	52.52%	0.5405	0.8939
Standalone SSIM Baseline	Video	58.14%	0.5576	0.8438
Baseline 3D-CNN + SSIM	Video	86.05%	0.8524	0.9123
Proposed Model + SSIM	Video	93.02%	0.9258	0.9655

*Seam Detection Recall: Standalone SSIM, Baseline+SSIM, and Proposed+SSIM all achieve 96.43% seam recall (27/28 detected) with 2.0 frames conditional median error. Neural fusion elevates authentic specificity from 40% (standalone SSIM) to 100% (proposed).

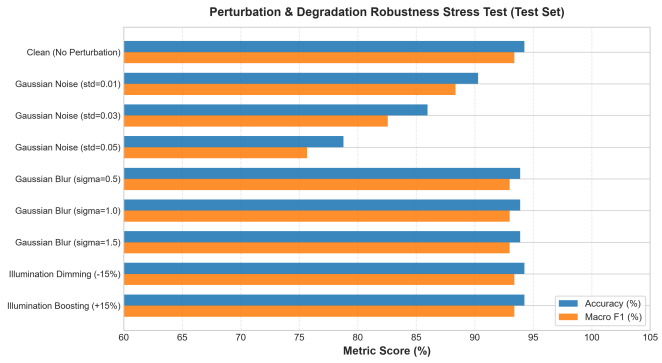


Fig. 3. Robustness stress tests under spatial box blur average pooling ($3 \times 3, 5 \times 5$), illumination scaling ($\pm 15\%$), and additive Gaussian noise ($\sigma \in \{0.01, 0.03, 0.05\}$).

accuracy degrades by only -0.36% (from 94.24% to 93.88%). Under illumination dimming (-15%) and boosting ($+15\%$), performance is 100% invariant (94.24% accuracy). Additive Gaussian tensor noise yields 90.29% accuracy at $\sigma = 0.01$ and 85.97% at $\sigma = 0.03$.

VII. Limitations and Methodological Caveats

1) Rapid Rotational Motion: Fast camera pans naturally induce frame-to-frame structural dissimilarity, producing secondary dips in SSIM trajectories that skew mean localization error to 34.39 frames, despite a median error of 2.0 frames.
2) Clip-Level Uncertainty vs. Video Independence: While bootstrap resampling ($N = 278$ clips) produces tight confidence intervals ([0.9690, 0.9929] OvR AUC), clips extracted from the same source video share background context. Generalization claims should be qualified regarding within-video statistical dependence.
3) Synthetic vs. Inpainted Forgery: Synthetic insertions introduce distinct donor boundaries. Real-world generative inpainting may incorporate temporal optical-flow blending that softens inter-frame structural seams.

VIII. Conclusion

We have presented an enhanced dual-stream R(2+1)D architecture with 3D-CBAM and Temporal Peak-Preserving Pooling (TP-Pool) for inter-frame video forgery detection. By concurrently modeling spatial texture and motion discontinuities, our framework resolves single-frame deletion seams and multi-frame insertion splices on a leak-free UCF-101 benchmark. Comprehensive comparative benchmarking confirms the failure of 2D frame pooling (53.24% accuracy) and recurrent temporal modeling (48.56% accuracy), highlighting the structural superiority of 3D factorized convolutions. Controlled matched-schedule ablations validate that TP-Pool provides an

unconfounded +11.51 percentage point gain in clip accuracy and +23.25 percentage points in video accuracy over global average pooling. Integrating deep network confidence with raw-frame structural similarity yields 93.02% video accuracy with zero authentic false alarms and 2.0 frames median localization error, establishing a transparent, reproducible benchmark for digital video forensics.

References

[1] R. Gowda and D. Pawar, "Deep learning-based forgery identification and localization in videos," Signal, Image and Video Processing, vol. 17, no. 4, pp. 1269–1277, 2023.

[2] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," IEEE Transactions on Image Processing, vol. 13, no. 4, pp. 600–612, 2004.

[3] D. Tran, H. Wang, L. Torresani, J. Ray, Y. LeCun, and M. Paluri, "A closer look at spatiotemporal convolutions for action recognition," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 6450–6459.

[4] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in Proceedings of the European Conference on Computer Vision (ECCV), 2018, pp. 3–19.

[5] K. Soomro, A. R. Zamir, and M. Shah, "UCF101: A dataset of 101 human actions classes from videos in the wild," arXiv preprint arXiv:1212.0402, 2012.

[6] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2017, pp. 2980–2988.